Istanbul Baraj Projesi - Guncel Durum ve Detayli Cikti Raporu

Kapsam ve Amac

Bu rapor, Istanbul baraj doluluk projesinin guncel durumunu, veri katmanlarini, model gelistirme turlarini, test sonuclarini, projeksiyon ciktilarini ve acik veri bosluklarini detayli sekilde ozetler.

Veri Katmanlari (Secili Ozetler)

- newdata_feature_store: na -> na (na ay)
- Resmi sehir suyu: 2010-01-01 -> 2023-12-01
- Reanalysis proxy: 1940-01-01 -> 2026-03-10
- Aylik operasyon proxy (transfer/NRW/geri kazan■m): na -> na

Model Gelistirme Turlari

Round 3 Ozet

family	model	rmse_pp	mape_pct	pearson_corr_pct	status
extra_param	hybrid_physics_stacked_exog	6.503846339208616	9.60215431087286	94.23514290816472	kept
extra_param	quantile_regressor_plus	7.781408402610931	9.859026669987475	91.4027488985479	candidate
extra_param	hybrid_physics_ensemble_phys_old	7.773890522819784	10.474449077252649	93.24176889774876	candidate
extra_param	water_balance_v4_corrected	8.703889909998326	11.551826571743634	94.43183531111052	candidate
extra_param	prophet_regressor_exog	7.966651483144593	11.981889035453278	92.01865454741468	candidate
water_balance_exog	water_balance_v5_exog	11.768243829381184	18.06199286469579	nan	rejected

Round 4 Ozet

artifact	status	rmse_pp	mape_pct	pearson_pct	note
monthly_operational_proxies	kept	nan	nan	nan	Aylik transfer/NRW/reclaimed proxy tablosu uretildi v
stacked_exog_ops	kept	6.580649091139963	9.523445322804566	94.21532642248675	Ops proxy sonrasi sonuc
quantile_ops_best	kept	7.781408402610931	9.859026669987475	91.4027488985479	Ops proxy sonrasi sonuc
prophet_ops_allproxy	rejected	17.96326300551316	24.921312563486534	61.87548387471165	Ops proxy sonrasi sonuc
water_balance_v6_ops_curve	rejected	11.363206918641266	15.409930326322003	92.3282992532975	Aylik transfer/NRW + alan-hacim egri vekili eklendi; p
prophet_subset_search_best	kept	7.966651483144593	11.981889035453278	92.01865454741468	En iyi Prophet hala src_rain_north tek ek regressor.
quantile_subset_search_best	kept	7.781408402610931	9.859026669987475	91.4027488985479	En iyi Quantile hala base boosted median.

Holdout Test Sonuclari (2016-2020)

model	rmse_pp	mae_pp	mape_pct	smape_pct	pearson_corr_pct	spearm
hybrid_physics_stacked_exog	6.580649091139963	5.311002662707452	9.523445322804566	10.043801978804352	94.21532642248675	95.5043
quantile_regressor_plus	7.781408402610931	5.9008026283590285	9.859026669987475	9.976279907648932	91.4027488985479	90.6751
hybrid_physics_ensemble_phys_old	7.773890522819784	6.000657070913325	10.474449077252649	9.828667957597132	93.24176889774876	93.4661
water_balance_v4_corrected	8.703889909998326	6.897682747074565	11.551826571743634	10.730825184658974	94.43183531111052	94.6033

prophet_regressor_exog	17.96326300551316	14.358897954591082	24.921312563486534	27.71533796937452	61.87548387471165	61.0502

2040 Sonu Baz Senaryo Uc Degerleri

model	pred_fill	pred_fill_pct
hybrid_physics_stacked_exog	0.1492568561494105	14.92568561494105
hybrid_physics_ensemble_phys_old	0.3578001102160605	35.78001102160605
water_balance_v4_corrected	0.029021234582677	2.9021234582677056
quantile_regressor_plus	0.4764826421385766	47.648264213857665
prophet_regressor_exog	0.391468949390547	39.1468949390547

MAPE 5 ve Agresif Aramalar

Kisa vadeli direct arama (MAPE 5 hedefi) ozeti

model	rmse_pp	mae_pp	mape_pct	smape_pct	pearson_corr_pct	spearman_cor
evo_direct_level_exog_blend	4.898806673277172	3.2729485823161952	5.617946651410206	5.610397597107901	96.43813167915712	95.548763545
gbr_d2_lr002	5.835730868647406	4.459084450438441	7.972740485690652	7.593334811226747	96.43813167915712	95.548763545
gbr_d3_lr0015	5.937048324227103	4.458547130020621	8.390930516034897	7.849258430117942	96.12766056588434	94.848569046
et_mid	5.887215651848968	4.720859389190134	9.11386601884326	8.448476570419615	96.57152450606982	95.520978049
et_deep	6.019703275706171	4.802070785316378	9.2834152990556	8.59882428254586	96.32190153214977	95.226451792
hgbr_mid	6.917888051989912	5.369229318895122	9.591472958121193	9.18850586629846	93.57293051836812	92.570158377
catboost_d4_lr003	6.103881647774073	4.795158555150639	9.909095907308489	8.955019336016942	97.68090869725783	96.732425673
catboost_d4_lr002	6.509189053352535	5.075015754072004	10.571272650410764	9.466119510775805	97.68502640670744	96.460127813
catboost_d5_lr002	6.852003459064616	5.4120823259857875	11.10418084642938	9.942092456956717	97.35709340062468	95.793275909
catboost_d6_lr015	7.904935784760711	6.261104291300097	12.930757983201197	11.401229874915678	96.42212554795806	94.248402333

Agresif direct-level arama ozeti

model	mode	rmse_pp	mape_pct	pearson_corr_pct	end_error_pp	weights
catboost_level_a	direct_level	6.204342679659403	10.062073495230472	97.5127604388305	12.316534648848492	nan
catboost_level_b	direct_level	6.003235172077312	9.558993219068284	97.07246023724696	10.744310328163149	nan
evo_direct_blend	direct_level_blend	4.878949043939169	5.481746572075834	96.6731540355139	1.8408565841451672	{"catboost_level_a": 0.0, "cat
gbr_level_a	direct_level	5.727779987972274	8.013201988604271	96.63255629134947	6.048832591906023	nan
gbr_level_b	direct_level	6.030107455426227	8.780824874669541	96.62668901386536	6.673767520344481	nan
et_level_a	direct_level	5.763626892306366	8.882627621218662	96.56908782922274	12.07747823416489	nan
hgbr_level_a	direct_level	6.5834195981574615	9.500651280536454	94.6022972659678	9.453398771012706	nan

Water-Balance Retry (Round 5)

model	rmse_pp	mape_pct	pearson_pct	base_2040_pct	decision
water_balance_v4_corrected	8.70389	11.551827	94.431835	2.902123	baseline_wb_best_so_far
water_balance_v6_ops_curve	11.363206918641266	15.409930326322003	92.3282992532975	6.213523530421872	reject
water_balance_v7_ops_residual	10.631368325062382	13.778137277006728	93.16868103216356	28.96952858814955	reject_but_less_bad_than_v6

Model Ilerleme Ozeti

family	old_model	new_model	old_mape_pct	new_mape_pct	mape_delta_pct	old_rmse
Stacked Hybrid	hybrid_physics_stacked_plus	hybrid_physics_stacked_exog	9.525550099014009	9.60215431087286	0.0766042118588519	7.090339
Prophet	prophet_regressor_plus	prophet_regressor_exog	13.001733381808132	11.981889035453278	-1.0198443463548532	10.664025
Quantile	quantile_regressor_plus	quantile_regressor_plus	9.859026669987475	9.859026669987475	0.0	7.781408

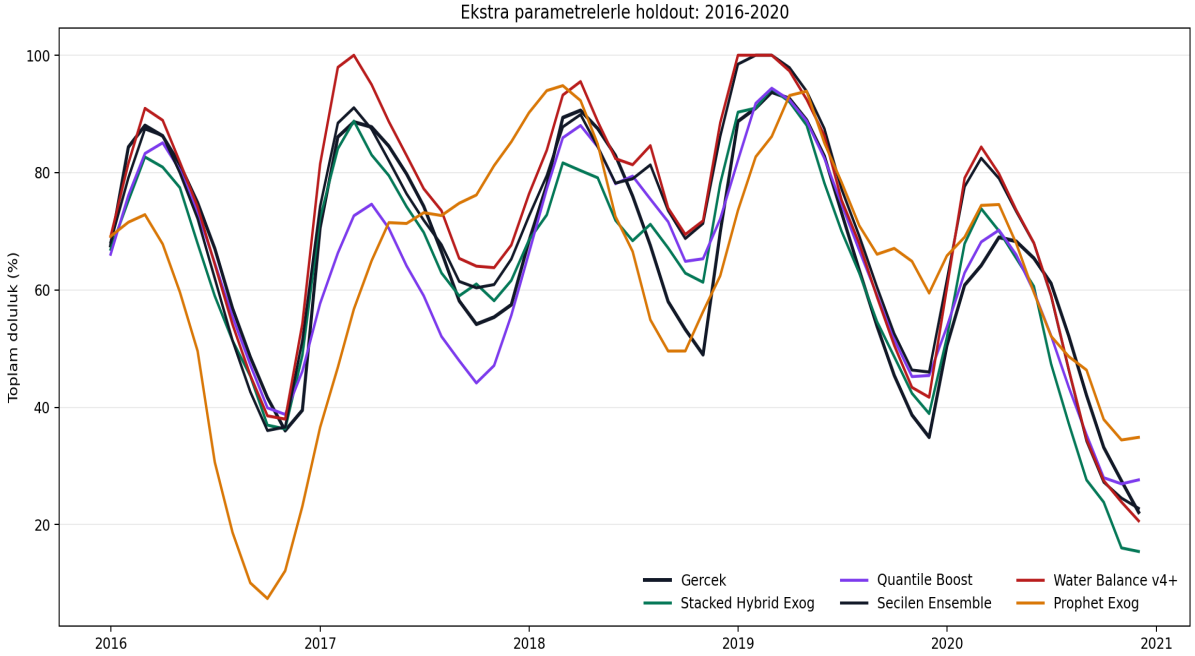
Ana Cikti Klasorleri

- output/final_delivery/istanbul_baraj_final_paket - output/istanbul_models_extra_params_2026_03_12 - output/istanbul_mape5_push_and_longhorizon_2026_03_12 - output/istanbul_aggressive_model_search_99_2026_03_12 - output/istanbul_preferred_projection_2040 -

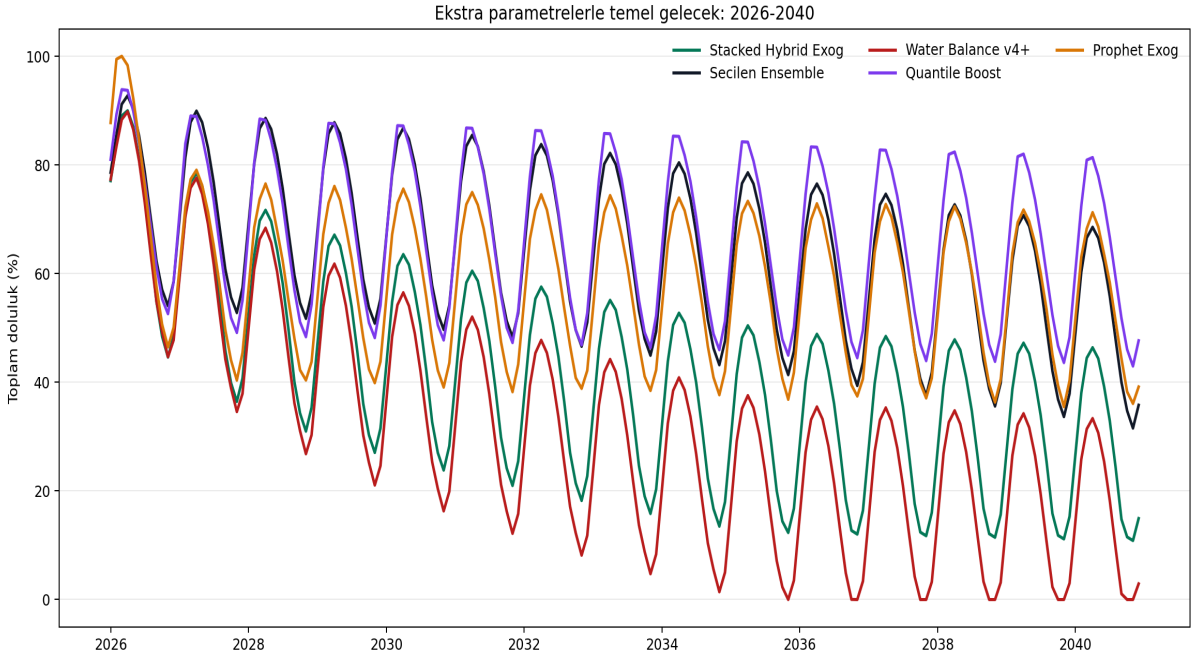
output/istanbul_preferred_operational_risk_2030

Gorseller

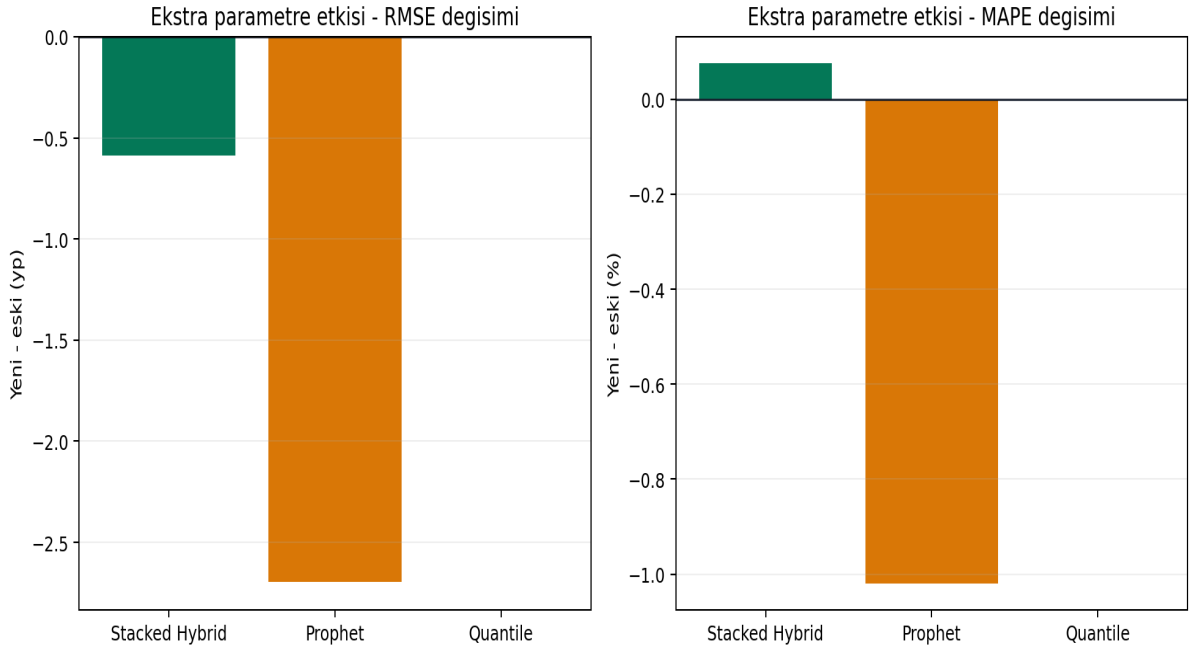
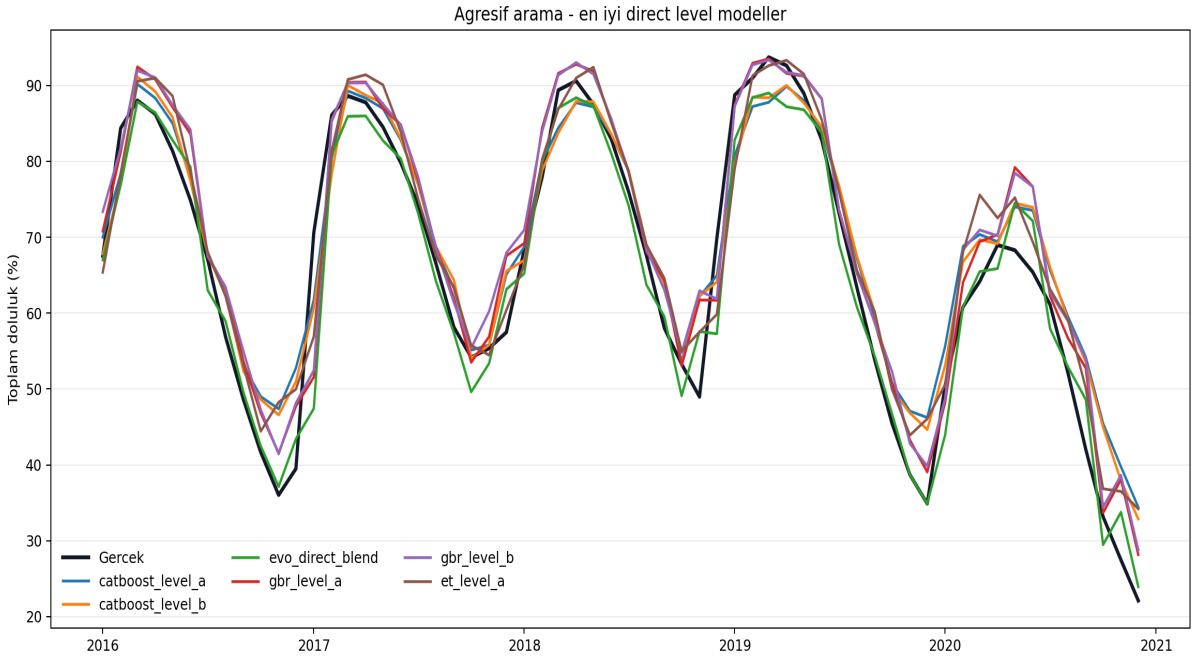
Gorsel: extra_param_models_holdout.png



Gorsel: extra_param_models_future.png



Gorsel: extra_param_improvement_vs_previous.png



Gorsel: aggressive_direct_level_top_models.png